Content

Learning objectives and content

Objectives

- To analyze and manipulate signals in both discrete and continuous time by applying fundamental transformations and determining key system properties such as causality, memory, linearity, and time-invariance.
- To characterize linear time invariant (LTI) systems using convolution and compute their responses to exponential and sinusoidal inputs in both continuous and discrete time.
- To apply continuous and discrete-time Fourier series and Fourier transforms to efficiently compute signal representations and solve real-world problems involving sampled signals.
- To write ordinary differential equations of LTI systems in state-space form and compute their transfer functions, as well as those of systems in cascade and feedback loops.
- To perform linear algebra operations such as computing the rank, null space, and column space of matrices, as well as computing eigenvalues and eigenvectors.
- To compute time and frequency responses of LTI systems in state-space form using coordinate transformations and the Laplace transform.
- To determine the poles and zeros of transfer functions, and sketch and interpret Bode plots.

Content
The course Signals and Systems covers the basics of signal and system analysis, adapted to the knowledge and demands of engineers, and includes the mathematical skills for handling matrices and matrix equations.

Topics that are covered in the signals part include convolution, Fourier series, Fourier transform, Discrete Fourier Transform and the Sampling Theorem Students learn that the representation of signals in frequency domain is just as important as the time domain representations they already know. In the systems part the focus lies on dynamical systems that convert input signals into output signals as is essential in many engineering applications and forms the basis of controller design. Different representations of systems are discussed including state-space models, transfer functions and frequency response functions (FRF), which will be connected to Fourier series, Fourier transform, Laplace transform of input and output signals.

The course begins with an introduction on the relevance of signal analysis and the use of frequency domain characterizations. The course starts by introducing some basic concepts of signals including convolution. Then it focuses on the decomposition of periodic signals into harmonic components via the Fourier series. The concepts are extended to general aperiodic signals, leading to the Fourier Transform. Additional topics such as the Discrete Fourier Transform and the Sampling Theorem are covered to enable students to use these concepts in real-life applications. This is the main focus of the project.

Based on this knowledge the step is then made towards system representations and analysis. Input/output (I/O) measurements form the basic starting point to obtain the Frequency Response Function (FRF) of a system. The FRF as a system representation is then connected to alternative system descriptions. The connection between FRF and state-space models via the transfer function is discussed in detail. The use of state-space models and transfer functions motivates a deeper understanding of mathematical analysis of matrix equations. This leads to a discussion of many important concepts including rank, eigenvalues, and eigenvectors of matrices, characteristic equation, the concept of basis (L5c) in R^n and coordinate transformations, similarity of matrices, diagonalization and (real) Jordan form. The connections between these mathematical notions and signal and system analysis will be clarified through various examples.

For Student Mobility Alliance students: This course requires knowledge of: This course requires knowledge of: - complex numbers, - differentiation; - integration; - solving systems of linear equations, i.e., solve Ax = b; - inverse of a matrix, determinant of a matrix; - linear combination of vectors; and - first-order differential equations. The lectures will

mostly be delivered in a hybrid format allowing you to follow online. The course assessment will be through a project and an exam with weights for the final grade of 25% and 75%, respectively. The project also requires group work. The final exam is a written on campus exam

2026

Signals and systems (4CA20)

Practical information

Course module
4CA20

Course type
Bachelor College

Credits (ECTS)
5 EC

Category
Deepening

Language of instruction
English

Offered by
Mechanical Engineering - Control Systems Technology

Short promotional description of the course
Students obtain knowledge and understanding of signals and systems in both the time and frequency domain. The course provides the essential basis for analyzing (controlled) dynamical systems and controller design in the follow-up courses

Course enrolment

OSIRIS Student
You can register yourself via OSIRIS Student. To do so, go to Enroll / Course.

Enrolment periods

- Block 4
  Timeslot(s)
  D - We 5-8, Tu 9-10, Fr 1-4

  Enrolment period
  15 November 2026 until 21 March 2027 23:59

  Unenrolment
  until 19 April 2027 23:59

Entrance requirements

Assumed previous knowledge
2WBB0 - Calculus (recommended) 4DA00 - Dynamics (recommended)

Entrance requirements course enrolment
You must meet the following requirements

- Completed none of the course modules listed below
  - Linear Algebra and Applications (2DBI00)
  - Vector Analysis (32VAN)
  - Signals and systems (4CB00)
  - Systems (5ESB0)
  - Linear Algebra and Applications (2DRR00)

Instructional modes

Instructional modes

- Guided selfstudy with notebook
  Remark
- Lecture
  Remark

Tests

Tests

- Written examination
  Final examination

  Test weight
  100

  Minimum grade

  Assessment
  First try: 80% exam + 20% project. Subsequent tries: 80% retake exam + 20% project Students may reuse the project marks, but only from the 2024-25 academic year onwards. For students retaking the course: 75% exam and 25% project if the project was done before year 26-27

Test enrolment

OSIRIS Student
You can register yourself via OSIRIS Student. To do so, go to Enroll / Test.

Written examination

- Block 4
  Enrolment period
  15 November 2026 until 6 June 2027 23:59

  Unenrolment
  until 6 June 2027

  Test date
  24 June 2027

- Block I
  Enrolment period
  15 November 2026 until 1 August 2027 23:59

  Unenrolment
  until 1 August 2027

  Test date
  19 August 2027

Lecturer(s)

Co-lecturer

- dr. R.A. González

Contactperson for the course

- P. Chanfreut Palacio

Responsible lecturer

- P. Chanfreut Palacio

Subject matter expert

- dr. R.A. González
- E. Shakhesi
- ir. T.M.J.T. Baltussen
- P. Chanfreut Palacio

Materials

Recommended material

- Via Canvas
- Alan Oppenheim (Author), Alan S. Willsky (Author), S.Hamid Nawab (Author)

  Signals and Systems

  2nd edition

  ISBN
  9781292025902
- David C. Lay (Author), Steven R. Lay (Author), Judi J. McDonald

  Linear Algebra and Its Applications

  ISBN
  9781292351216

- Gene F. Franklin , J. David Powell , Abbas Emami-Naeini

  Feedback Control of Dynamic Systems

  ISBN
  9781292274522

- João P. Hespanha

  Linear Systems Theory

  ISBN
  9780691179575

Do you want to know more?

P. Chanfreut Palacio can tell you more about this course.
Send an e-mail to p.chanfreut.palacio@tue.nl